WELLNESS & AI PLATFORM

Personal Wellness Tracking & Machine Learning Lab

An end-to-end intelligence suite for personal health check-ins, feature importance modeling, and predictive what-if simulations.

MODEL PRECISION	CORE ENGINE	KEY DRIVERS
93% Accuracy	Random Forest ML	Stress & Energy

ARCHITECTURE OVERVIEW

Core Platform Modules

Dashboard

Real-time timeline tracking mood, energy, sleep, and stress.

Daily Check-in

Structured self-reporting for energy, social support, and sleep.

AI / ML Lab

Model precision benchmarking and feature importance evaluation.

Pattern Explorer

Interactive what-if simulator to forecast wellness outcomes.

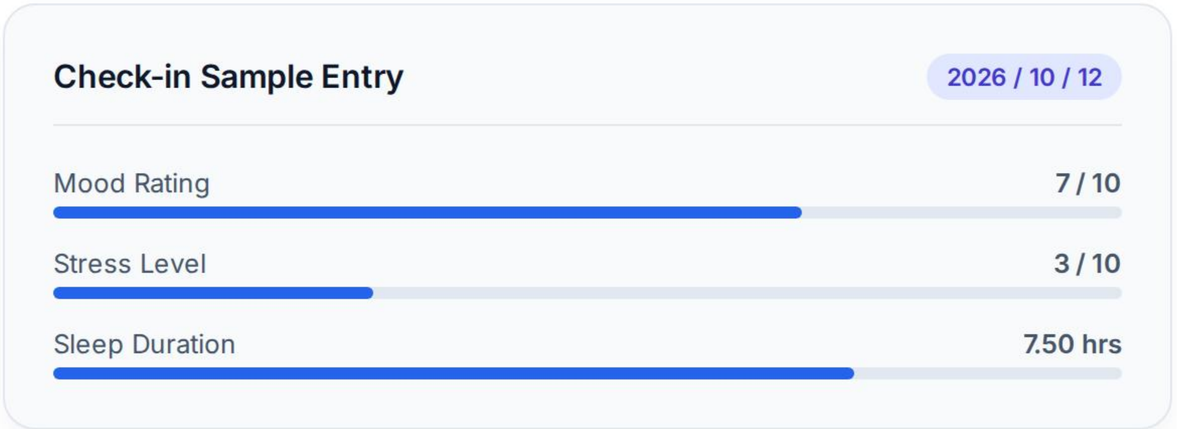
DATA COLLECTION

Daily Wellness Check-in Flow

1. **Core Metrics Rating**
   Sliders capturing Mood (1-10), Stress (1-10), and Energy (1-10).

2. **Sleep & Social Support**
   Duration entry (e.g., 7.50 hrs) alongside perceived social connectedness.

3. **Contextual Tagging**
   Activity logging (e.g., Exercise) with optional private journal notes.



MODEL BENCHMARKING

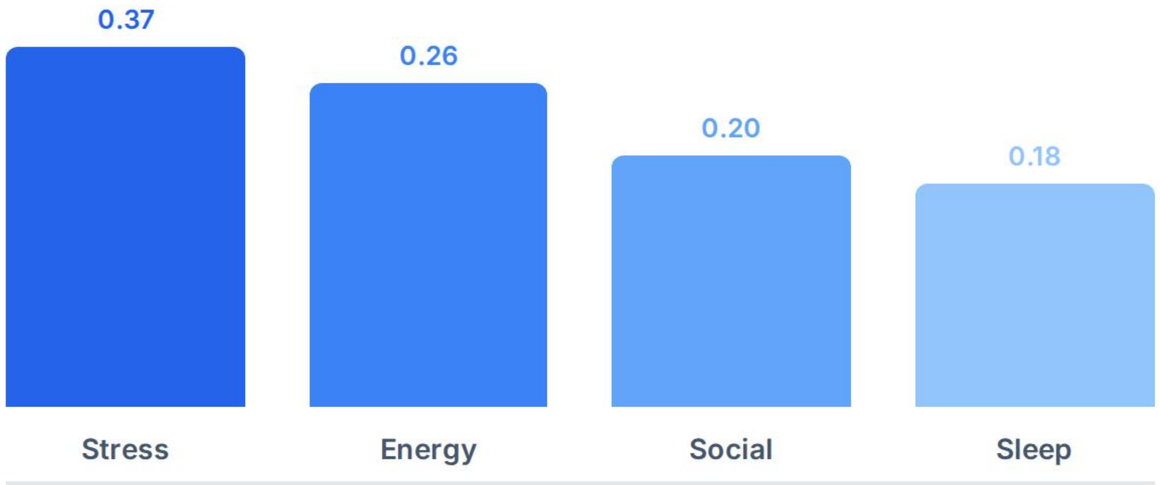
Random Forest Model Evaluation

Accuracy	Precision	Recall	F1-Score
0.93	0.93	0.93	0.93

Educational Model Note: Trained on synthetic wellness records. Demonstrates high classification balanced across metrics for predictive wellness prototyping.

FEATURE ANALYTICS

Key Drivers of Wellness Classification



Model Observations

- **Stress Level (37%)** serves as the highest weighted predictor for personal wellness status.
- **Energy Ratings (26%)** represent the secondary impact variable in daily model outputs.
- **Social Support & Sleep** combined contribute 38% total weight toward predictive scoring.

PREDICTIVE TOOLING

What-If Wellness Simulator

Hypothetical Sleep (Hours)

7.50 hrs

Hypothetical Energy (1–10)

7 / 10

Hypothetical Stress (1–10)

3 / 10

Hypothetical Social Connection (1–10)

7 / 10

Run What-If Simulation Engine

GOVERNANCE & OPS

Admin Portal & Security Audit Logs

Registered Users	Total Check-ins	Active Experiments
2	4	0

ID	Created At	User ID	Action	Details
1	2026-10-12 12:04:54	None	VIEW_ADMIN_DASHBOARD	Viewed system logs
2	2026-10-12 12:13:12	None	VIEW_ADMIN_DASHBOARD	Viewed system logs
3	2026-10-12 12:24:33	None	VIEW_ADMIN_DASHBOARD	Viewed system logs

Wellness Analytics Platform

Empowering individual wellbeing with machine learning precision and actionable insights.